

	DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Supplementary End Semester Examination – Winter 2023 Course: B. Tech. Branch: All Semester: II Subject Code & Name: Engineering Physics (BTBS202P) Max Marks: 60 Date:17/01/2024 Duration: 3 Hr.			
	Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks	
Q. 1	Solve Any Two of the following.		12	
A)	Define forced Vibrations. Derive differential equation for forced vibrations.	(CO1) (Remember & understand)	6	
B)	Explain the construction, working for production of ultrasonic waves using Piezoelectric oscillator.	(CO1) (Understand)	6	
C)	Derive differential equation for wave motion.	(CO1) (Understand)	6	
Q.2	Solve Any Two of the following.		12	
A)	In Newton's rings, derive an expression for diameter of n^{th} bright ring and dark ring.	(CO2) (Understand)	6	
B)	Explain the terms 1) Absorption 2) Spontaneous emission 3) Stimulated Emission 4) Population Inversion	(CO2) (Understand)	6	
C)	Explain Huygen's theory of double refraction. A glass plate is to be used as a polarizer. Find the angle of polarization for it. Also find the angle of refraction. ($\mu = 1.54$)	(CO2) (Understand)	6	
Q. 3	Solve Any Two of the following.		12	
A)	Explain the motion of electron in parallel and perpendicular magnetic field.	(CO3) (Understand)	6	
B)	Write short note on Geiger Muller Counter.	(CO3) (Understand)	6	

C)	Derive Schrodinger's time dependent wave equation.	(CO3) (Understand)	6
Q.4	Solve the following questions.		12
A)	Define Atomic Radius, Coordination Number and Atomic Packing Fraction. Derive the relation between lattice parameter 'a' and crystal density 'ρ'.	(CO4) (Understand)	6
B)	Explain Characteristics X-ray spectra. Calculate the wavelength of X-rays when a potential difference of 20 KV is applied between filament and anode.	(CO4) (Understand)	6
Q. 5	Solve Any Two of the following.		12
A)	Explain Diamagnetic, Paramagnetic and Ferromagnetic materials with examples and diagram.	(Understand)	6
B)	Distinguish between Type I and Type II superconductors.	(Understand)	6
C)	Derive an expression for conductivity of Intrinsic and extrinsic (P Type & N Type) Semiconductors.	(Understand)	6
	*** End ***		